

Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo + Discrepancy

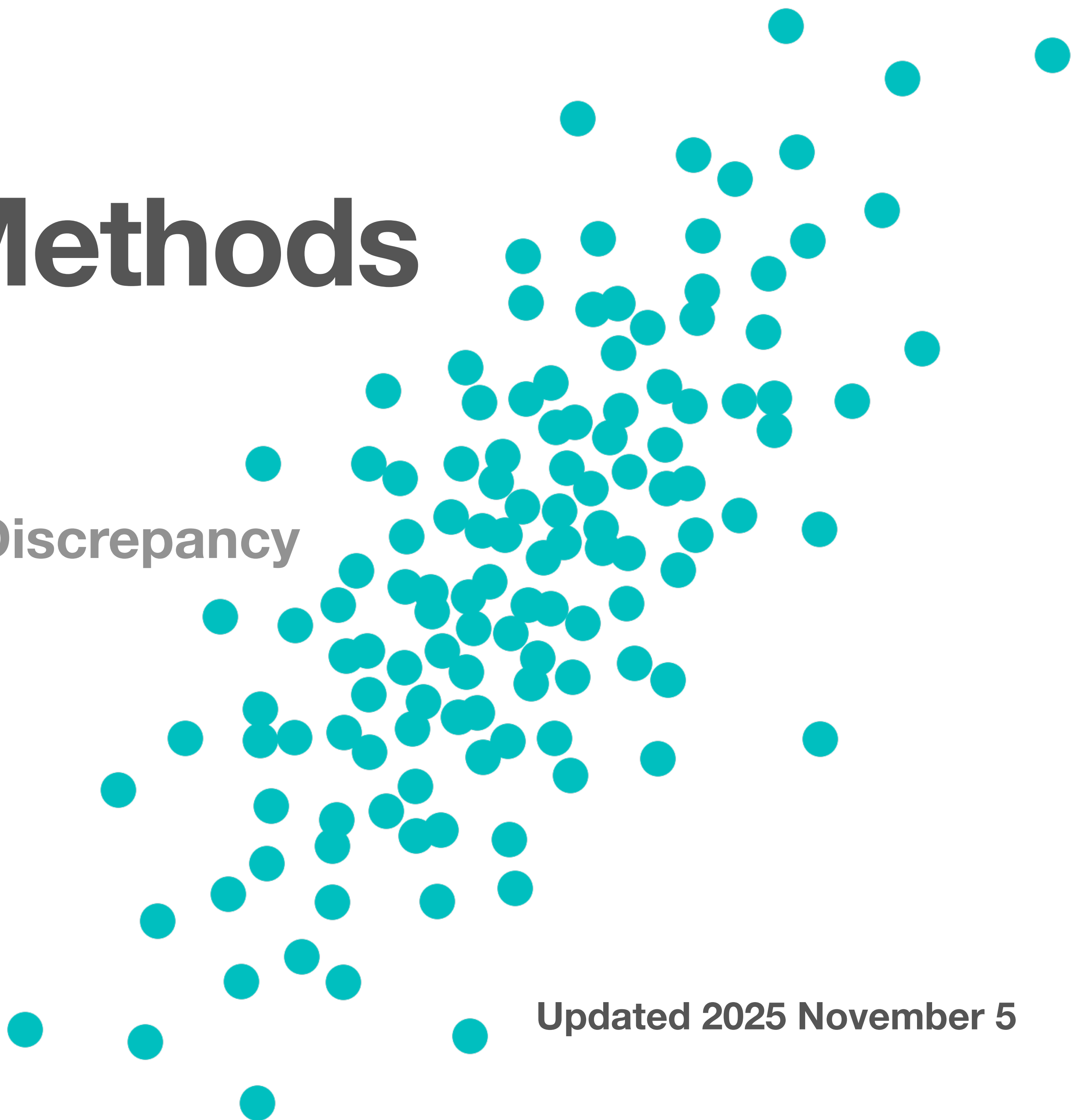
Improving Efficiency

Selected Topics

Git website and repository
Canvas

Fred Hickernell, Fall 2025

Updated 2025 November 5

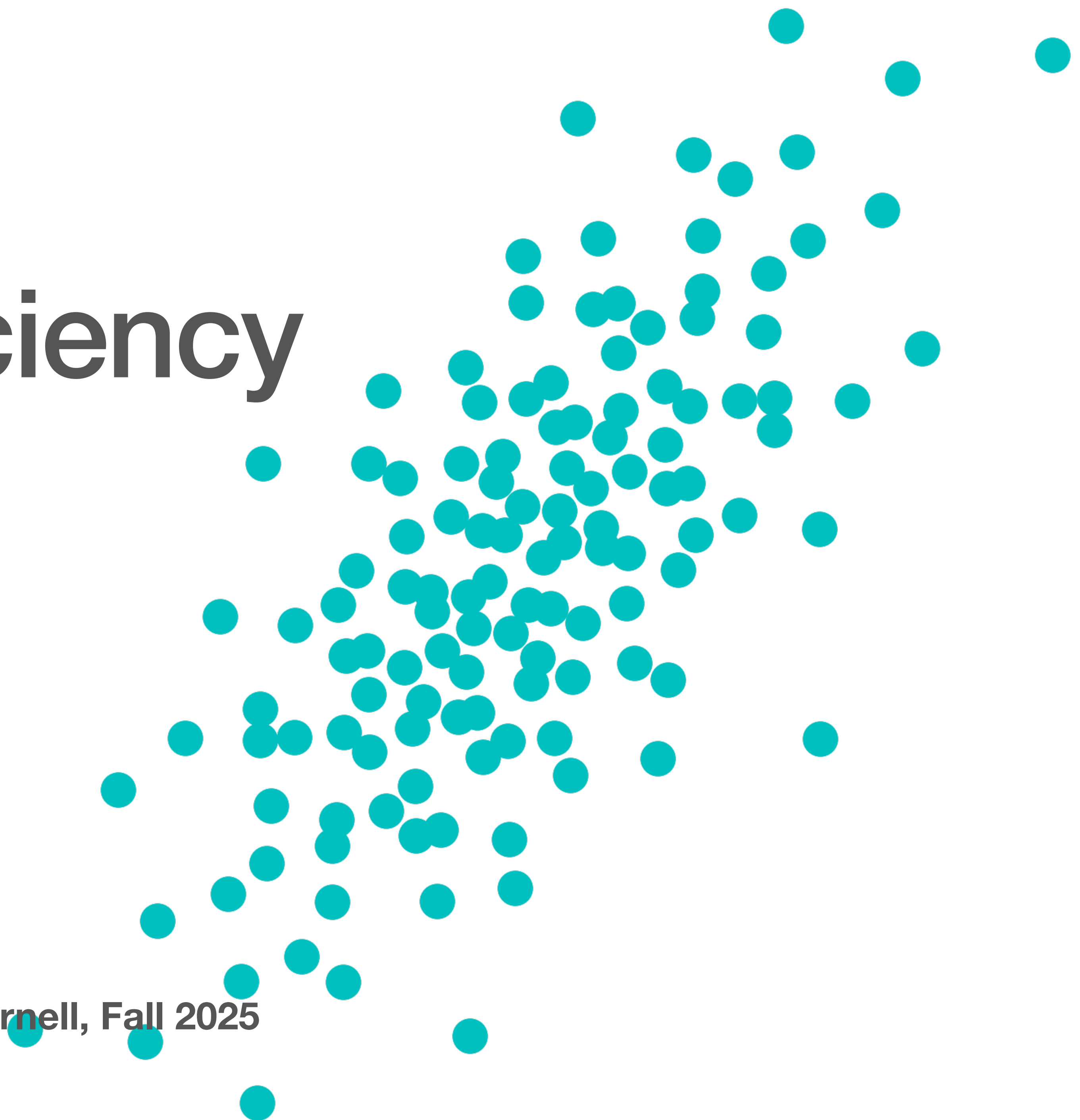


Improving Efficiency

Owen, Chapters 8–10, 15–17

Test 2 Oct 29

MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025





Variance Reduction

$$\mu = \mathbb{E}[f(X)] = \mathbb{E}[g(X)], \quad X \sim \mathcal{U}[0,1]^d$$

Is it better to use

$$\hat{\mu}_{f,n} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \quad \text{or} \quad \hat{\mu}_{g,n} = \frac{1}{n} \sum_{i=0}^{n-1} g(X_i), \quad X_0, X_1, \dots \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}[0,1]^d$$

to approximate μ ? Since both sample means are unbiased, we can compare their variances (mean squared errors):

$$\text{var}(\hat{\mu}_{f,n}) = \frac{\text{var}(f(X))}{n} \quad \text{var}(\hat{\mu}_{g,n}) = \frac{\text{var}(g(X))}{n}$$

so it depends on the variances of the two functions



Keister example

$$\mu = \int_{\mathbb{R}^d} \cos(\|t\|) \exp(-\|t\|^2) dt$$

$$\mathbf{x} = (\Phi(at_1), \dots, \Phi(at_d)), \quad \Phi \text{ is the CDF of the standard normal}$$

$$d\mathbf{x} = a^d \phi(at_1) \cdots \phi(at_d) dt = \frac{a^d \exp(-a^2 \|t\|^2/2)}{(2\pi)^{d/2}} dt$$

$$t = (\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d)) / a$$

$$\begin{aligned} \mu = \int_{[0,1]^d} & \frac{(2\pi)^{d/2}}{a^d} \cos\left(\left\|(\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d))\right\| / a\right) \\ & \times \exp\left(-\frac{2-a^2}{2a^2} \left\|(\Phi^{-1}(x_1), \dots, \Phi^{-1}(x_d))\right\|^2\right) d\mathbf{x}, \quad (a^2 \leq 2) \end{aligned}$$

Whole family $f(\mathbf{x}; a)$ for which $\mu = \mathbb{E}[f(X; a)]$ for $X \sim \mathcal{U}[0,1]^d$



Variable transformations

Suppose that you want to approximate

$$\mu = \int_{\mathcal{T}} g(\mathbf{t}) \, d\mathbf{t}$$

by Monte Carlo. You need to write it as an expectation. Let $\mathbf{t} = \boldsymbol{\psi}(\mathbf{x})$, for some **invertible transformation**, $\boldsymbol{\psi} : \mathcal{X} \rightarrow \mathcal{T}$. Define the **Jacobian determinant** as

$$J(\mathbf{x}; \boldsymbol{\psi}) := \det \left(\left(\frac{\partial \psi_i}{\partial x_j} \right)_{i,j=1}^d \right), \quad J(\mathbf{t}; \boldsymbol{\psi}^{-1}) := \det \left(\left(\frac{\partial \psi_i^{-1}}{\partial t_j} \right)_{i,j=1}^d \right), \quad J(\boldsymbol{\psi}(\mathbf{x}); \boldsymbol{\psi}^{-1}) = \frac{1}{J(\mathbf{x}; \boldsymbol{\psi})}$$

Then

$$\mu = \int_{\mathcal{X}} g(\boldsymbol{\psi}(\mathbf{x})) |J(\mathbf{x}; \boldsymbol{\psi})| \, d\mathbf{x} = \int_{\mathcal{X}} \underbrace{\frac{g(\boldsymbol{\psi}(\mathbf{x}))}{\varrho_{\mathbf{X}}(\mathbf{x})} |J(\mathbf{x}; \boldsymbol{\psi})|}_{f(\mathbf{x})} \varrho_{\mathbf{X}}(\mathbf{x}) \, d\mathbf{x} = \mathbb{E}_{\mathbf{X} \sim \varrho_{\mathbf{X}}} [f(\mathbf{X})]$$



Variable transformations

$$\mu = \int_{\mathcal{T}} g(t) \, dt = ?$$

Let $t = \psi(x)$, for some **invertible transformation**, $\psi : \mathcal{X} \rightarrow \mathcal{T}$. Then

$$\mu = \int_{\mathcal{X}} g(\psi(x)) |J(x; \psi)| \, dx = \int_{\mathcal{X}} \underbrace{\frac{g(\psi(x))}{\varrho(x)} |J(x; \psi)|}_{f(x)} \varrho(x) \, dx = \mathbb{E}_{\mathbf{X} \sim \varrho}[f(\mathbf{X})]$$

If $\mathbf{X} \sim \mathcal{U}[0,1]^d$ (because we know how generate **uniform** samples), then

$$\mu = \int_{[0,1]^d} \underbrace{g(\psi(x)) |J(x; \psi)|}_{f(x)} \, dx = \mathbb{E}_{\mathbf{X} \sim \mathcal{U}[0,1]^d}[f(\mathbf{X})]$$



Random variable transformations

Let

- T be a random variable with sample space $\mathcal{T}$ and density ϱ_T
- X be a random variable with sample space $\mathcal{X}$ and density ϱ_X
- $T = \psi(X)$, for some **invertible transformation**, $\psi : \mathcal{X} \rightarrow \mathcal{T}$. Then

$$\varrho_T(t) = \varrho_X(\psi^{-1}(t)) |J(t; \psi^{-1})|, \quad \varrho_T(\psi(x)) = \varrho_X(x) |J(\psi(x)); \psi^{-1})| = \frac{\varrho_X(x)}{|J(x; \psi)|}$$

$$\mu = \int_{\mathcal{T}} \underbrace{h(t) \varrho_T(t)}_{g(t)} dt = \mathbb{E}_{T \sim \varrho_T}[h(T)] = \mathbb{E}_{X \sim \varrho_X}[h(\psi(X))] = \int_{\mathcal{X}} h(\psi(x)) \varrho_X(x) dx$$

$$\text{or } = \int_{\mathcal{X}} \frac{h(\psi(x)) \varrho_T(\psi(x))}{\varrho_X(x)} |J(x; \psi)| \varrho_X(x) dx = \int_{\mathcal{X}} h(\psi(x)) \varrho_X(x) dx$$



Importance sampling

Suppose that you want to approximate

$$\mu = \mathbb{E}_{\mathbf{T} \sim \varrho_{\mathbf{T}}} [h(\mathbf{T})] = \int_{\mathcal{T}} h(\mathbf{t}) \varrho_{\mathbf{T}}(\mathbf{t}) \, d\mathbf{t} = \int_{\mathcal{T}} \underbrace{h(\mathbf{t}) \frac{\varrho_{\mathbf{T}}(\mathbf{t})}{\varrho_{\mathbf{X}}(\mathbf{t})}}_{f(\mathbf{t})} \overbrace{\varrho_{\mathbf{X}}(\mathbf{t})}^{\text{likelihood ratio}} \, d\mathbf{t} = \mathbb{E}_{\mathbf{X} \sim \varrho_{\mathbf{X}}} [f(\mathbf{X})]$$

We try to choose $\varrho_{\mathbf{X}}$ to make $\text{var}[f(\mathbf{X})]$ small when doing IID sampling. If $\mathbf{T} = \boldsymbol{\psi}(\mathbf{X})$ for $\mathbf{X} \sim \mathcal{X}[0,1]^d$ —since often find generate non-uniform samples from uniform ones—then

$$\mu = \int_{\mathcal{T}} h(\mathbf{t}) \varrho_{\mathbf{T}}(\mathbf{t}) \, d\mathbf{t} = \mathbb{E}_{\mathbf{T} \sim \varrho_{\mathbf{T}}} [h(\mathbf{T})] = \mathbb{E}_{\mathbf{X} \sim \mathcal{U}[0,1]^d} [h(\boldsymbol{\psi}(\mathbf{X}))] = \int_{[0,1]^d} h(\boldsymbol{\psi}(\mathbf{x})) \, d\mathbf{x}$$



Ex. Brownian motion with drift

In computational finance certain outcomes, such as positive option payoffs, may be **rare**

- Adding a drift to the Brownian motion **increases** the probability that asset paths yield positive payoffs and
- **Reduces the variance** of our estimator of the option price (population mean)
- Paths with positive payoffs receive **smaller sample weights** to compensate and keep the mean the same

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \varrho(\mathbf{x}) d\mathbf{x}, \quad \varrho(\mathbf{x}) = \frac{\exp(-\mathbf{x}^\top \Sigma^{-1} \mathbf{x} / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \Sigma = \left(\min(t_i, t_j) \right)_{i,j=1}^d$$

Here the density ϱ is a discretized **Brownian motion**



Ex. Brownian motion with drift cont'd

$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \varrho(\mathbf{x}) d\mathbf{x}$$
$$\varrho(\mathbf{x}) = \frac{\exp(-\mathbf{x}^\top \Sigma^{-1} \mathbf{x} / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \Sigma = \left(\min(t_i, t_j) \right)_{i,j=1}^d$$

We importance sample:

$$\varrho_{\text{drift}}(\mathbf{x}) = \frac{\exp(-(\mathbf{x} - \mathbf{a})^\top \Sigma^{-1} (\mathbf{x} - \mathbf{a}) / 2)}{\sqrt{(2\pi)^d |\Sigma|}}, \quad \mathbf{a} = \alpha(t_1, \dots, t_d)^\top$$

$$\text{likelihood ratio} = \frac{\varrho(\mathbf{x})}{\varrho_{\text{drift}}(\mathbf{x})} = \exp(-\alpha x_d + \alpha^2 t_d / 2)$$

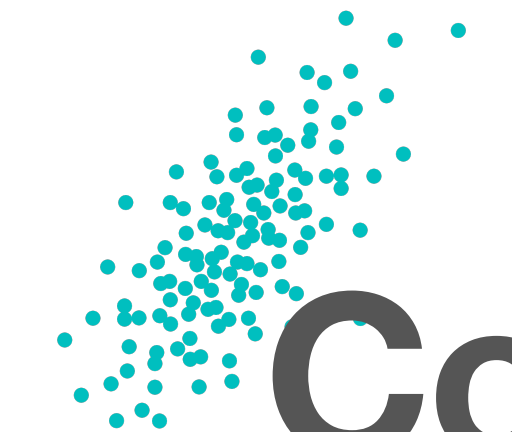
$$\text{price} = \int_{\mathbb{R}^d} \text{payoff}(\mathbf{x}) \exp(-\alpha x_d + \alpha^2 t_d / 2) \varrho_{\text{drift}}(\mathbf{x}) d\mathbf{x}$$

Note that the weight gets **smaller** as αx_d increases or $\alpha^2 t_d$ decreases



Variable transformation / importance sampling observations

- Variable transformation and importance sampling are **intertwined**
- The **choice** of sampling density or variable transformation affects the variance of the sample mean
- You must choose the variable transformation so that
 - The Jacobian determinant, $J(\mathbf{x}; \boldsymbol{\psi})$, is **finite**
 - The probability density, $q_{\mathbf{X}}(\mathbf{x})$, does not **vanish** unless integrand vanishes
- You might try **pilot** samples to help choose transformation



Control variates

$$\mu = \mathbb{E}[f(X)] \iff \mu = \mathbb{E}[f(X) - \beta_1 \eta_1(X) - \dots - \beta_m \eta_m(X)]$$

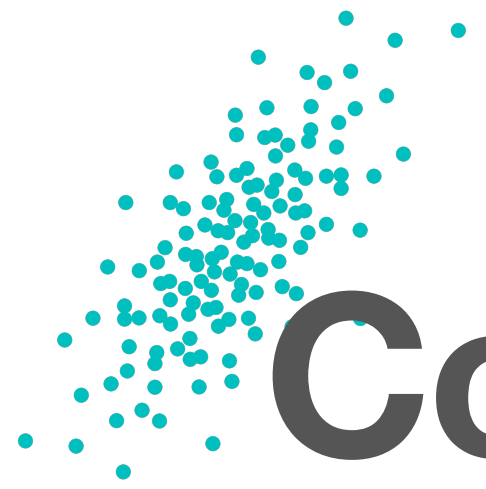
provided that $\mathbb{E}[\eta_1(X)] = \dots \mathbb{E}[\eta_m(X)] = 0$. Want to arrange such that

$$\text{var}[f(X) - \beta_1 \eta_1(X) - \dots - \beta_m \eta_m(X)] \leq \text{var}[f(X)]$$

Then

$$\hat{\mu}_{\text{CV},n} = \frac{1}{n} \sum_{i=0}^{n-1} [f(X_i) - \beta_1 \eta_1(X_i) - \dots - \beta_m \eta_m(X_i)] \text{ converges faster than } \hat{\mu}_n = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i)$$

How do we **minimize** $\text{var}[f(X) - \beta_1 \eta_1(X) - \dots - \beta_m \eta_m(X)]$?



Control variates (cont'd)

$$\hat{\mu}_{\text{CV},n} = \frac{1}{n} \sum_{i=0}^{n-1} [f(X_i) - \beta_1 \eta_1(X_i) - \cdots - \beta_m \eta_m(X_i)]$$

How do we **minimize** $\text{var}[f(X) - \beta_1 \eta_1(X) - \cdots - \beta_m \eta_m(X)]$? Use least squares regression with centered response and explanatory variables to choose $\hat{\beta}$:

$$\hat{\beta} = \underset{\mathbf{b}}{\text{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2$$

$$\text{where } \mathbf{y} = \begin{pmatrix} f(\mathbf{x}_0) - \hat{\mu}_n \\ \vdots \\ f(\mathbf{x}_{n-1}) - \hat{\mu}_n \end{pmatrix}, \quad \mathbf{H} = \begin{pmatrix} \eta_1(\mathbf{x}_0) - \hat{\mu}_{\eta_1,n} & \cdots & \eta_m(\mathbf{x}_0) - \hat{\mu}_{\eta_m,n} \\ \vdots & & \vdots \\ \eta_1(\mathbf{x}_{n-1}) - \hat{\mu}_{\eta_1,n} & \cdots & \eta_m(\mathbf{x}_{n-1}) - \hat{\mu}_{\eta_m,n} \end{pmatrix}$$

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{x}_i), \quad \hat{\mu}_{\eta_1,n} = \frac{1}{n} \sum_{i=0}^{n-1} \eta_1(\mathbf{x}_i), \quad \dots, \quad \hat{\mu}_{\eta_m,n} = \frac{1}{n} \sum_{i=0}^{n-1} \eta_m(\mathbf{x}_i)$$



Least Squares

How do we solve

$$\hat{\beta} = \underset{\mathbf{b}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2 = \underset{\mathbf{b}}{\operatorname{argmin}} (\mathbf{y} - \mathbf{H}\mathbf{b})^T (\mathbf{y} - \mathbf{H}\mathbf{b})$$

Let

$$\mathbf{b} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{y} + \mathbf{c}$$

Then

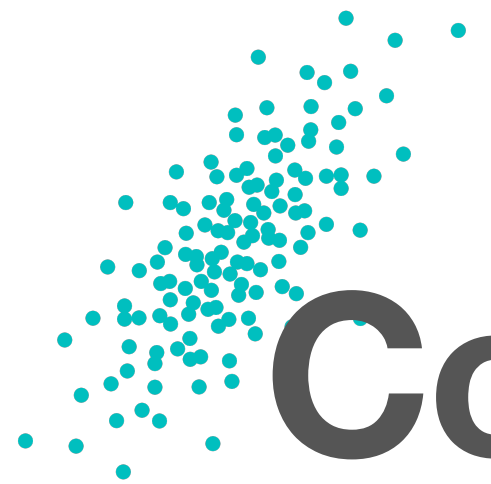
$$\begin{aligned} \|\mathbf{y} - \mathbf{H}\mathbf{b}\|^2 &= \|\mathbf{y} - \mathbf{H}[(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{y} + \mathbf{c}]\|^2 \\ &= \|[\mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T] \mathbf{y} + \mathbf{H}\mathbf{c}\|^2 \\ &= ([\mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T] \mathbf{y} + \mathbf{H}\mathbf{c})^T ([\mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T] \mathbf{y} + \mathbf{H}\mathbf{c}) \\ &= \mathbf{y}^T [\mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T] \mathbf{y} + \mathbf{c}^T \mathbf{H}^T \mathbf{H} \mathbf{c} \end{aligned}$$

So the optimum comes by setting $\mathbf{c} = \mathbf{0}$



Control variates

- Should be **clever** in choosing the η_j
 - **Good** choices can decrease the sample size required
 - **Bad** choices only hurt in terms of increased **computation** time, but not increased sample size
- The β_j should be chosen by **least squares** for IID sampling
- For **low discrepancy** sequences, the β_j should be chosen to reduce the **variation** (not variance)



Conditional Monte Carlo

Recall that

$$\mathbb{E}(Y) = \mathbb{E}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})], \quad \text{var}(Y) = \mathbb{E}_{\mathbf{Z}}[\text{var}(Y | \mathbf{Z})] + \text{var}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})]$$

Which implies that

$$\text{var}_{\mathbf{Z}}[\mathbb{E}_Y(Y | \mathbf{Z})] \leq \text{var}(Y), \quad \text{var} \left(\frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_Y(Y | \mathbf{Z}_i) \right) \leq \text{var} \left(\frac{1}{n} \sum_{i=0}^{n-1} Y_i \right)$$

Let $(Y = f(\mathbf{X}_{1:d}), \mathbf{X}_{1:d})$ be a vector of random variables

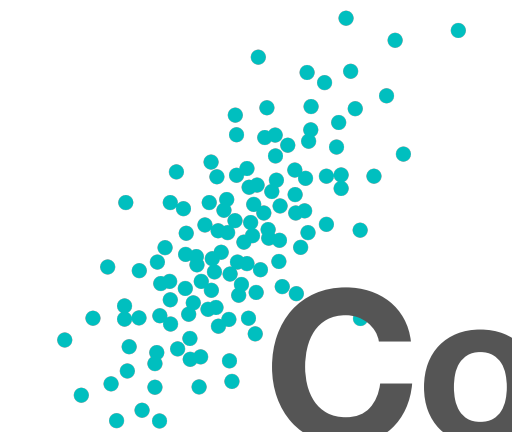
Let $\mathbf{Z} = \mathbf{X}_{2:d}$

If $g(\mathbf{X}_{2:d}) = \mathbb{E}_Y(Y | \mathbf{X}_{2:d})$ has an analytic form, then

$$\mu = \mathbb{E}(Y) = \mathbb{E}_{\mathbf{X}_{2:d}}[g(\mathbf{X}_{2:d})] \text{ and } \text{var}(Y) \geq \text{var}_{\mathbf{X}_{2:d}}[g(\mathbf{X}_{2:d})]$$

If $h(y, \mathbf{X}_{2:d}) = \varrho_{Y|\mathbf{X}_{2:d}}(y | \mathbf{x}_{2:d})$ has an analytic form, then

$$\varrho_Y(y) = \mathbb{E}_{\mathbf{X}_{2:d}}[h(y, \mathbf{X}_{2:d})] \quad \text{probability density expressed as expectation}$$



Conditional Monte Carlo for Density Estimation

Estimate the probability density, ϱ , for $Y = f(X)$, where $X \sim \mathcal{U}[0,1]^d$

If one can identify an **increasing** g , such that

$$y = f(\mathbf{x}) \iff x_1 = g(y; \mathbf{x}_{2:d})$$

Then

$$\begin{aligned} F_{Y|X_{2:d}}(y | \mathbf{x}_{2:d}) &= \mathbb{P}[Y \leq y | X_{2:d} = \mathbf{x}_{2:d}] = \mathbb{P}[f(X) \leq y | X_{2:d} = \mathbf{x}_{2:d}] \\ &= \mathbb{P}[X_1 \leq g(y; \mathbf{x}_{2:d})] = g(y; \mathbf{x}_{2:d}) \end{aligned}$$

$$\varrho(y) = \mathbb{E}[\varrho_{Y|X_{2:d}}(y | \mathbf{x}_{2:d})] = \int_{[0,1]^{d-1}} g'(y; \mathbf{x}) \, d\mathbf{x}$$



Ex. Density of a sum of uniforms

Let $Y = \gamma_1 X_1 + \cdots + \gamma_d X_d$, where $X \sim \mathcal{U}[0,1]^d$. What is the PDF of Y ?

$$g(y; x_2, \dots, x_d) = \frac{1}{\gamma_1} (y - \gamma_2 x_2 - \cdots - \gamma_d x_d), \quad \gamma_2 x_2 + \cdots + \gamma_d x_d \leq y \leq \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_d x_d$$

$$g'(y; x_2, \dots, x_d) = \frac{1}{\gamma_1}, \quad \gamma_2 x_2 + \cdots + \gamma_d x_d \leq y \leq \gamma_1 + \gamma_2 x_2 + \cdots + \gamma_d x_d$$



Antithetic Sampling

If you are sampling from a probability distribution with symmetry, then you can evaluate f at both X_i and $T(X_i)$, where $T(X)$ has the same distribution as X , e.g.,

$$T(X) = \mathbf{1} - X \text{ for } X \sim \mathcal{U}[0,1]^d \text{ or } T(X) = -X \text{ for } X \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

Note that if X_i are IID, then X_i and $T(X_i)$ are not independent, but $(X_i, T(X_i))$ are

When does $\hat{\mu}_{\text{AS},n} = \frac{1}{2n} \sum_{i=0}^{n-1} [f(X_i) + f(T(X_i))]$ have a smaller variance than

$$\hat{\mu}_{2n} = \frac{1}{2n} \sum_{i=0}^{2n-1} f(X_i) \text{ for IID } X_i?$$



Antithetic Sampling

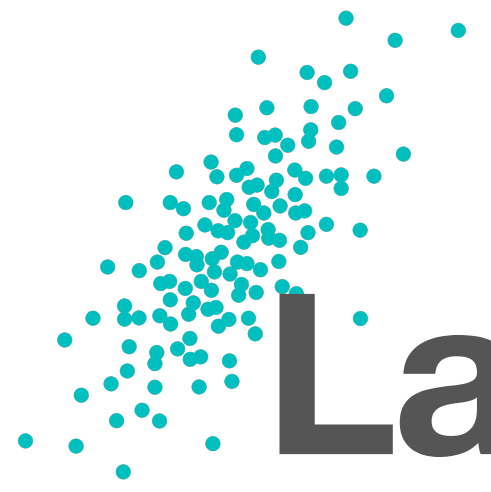
Note that

$$\text{var}(\hat{\mu}_{\text{AS},n}) = \frac{1}{4n^2} \sum_{i=0}^{n-1} \text{var}(f(X_i) + f(T(X_i))) = \frac{\text{var}(f(X))}{2n} [1 + \text{corr}(f(X), f(T(X)))]$$

Whereas

$$\text{var}(\hat{\mu}_{2n}) = \frac{\text{var}(f(X))}{2n}$$

So antithetica sampling helps if $\text{corr}(f(X), f(T(X))) < 0$



Latin Hypercube Sampling in One Dimension

Let's be intentional about placing uniform looking points on $[0,1]$

$$X_i = (i + U_i)/n, \quad U_0, U_1, \dots \stackrel{\text{IID}}{\sim} \mathcal{U}[0,1], \quad i = 0, \dots, n-1, \quad \hat{\mu}_{\text{LHS},n} = \frac{1}{n} \sum_{i=0}^{n-1} f(X_i)$$

Note that $X_i = (i + U_i)/n \sim \mathcal{U}[i/n, (i+1)/n]$, and so

$$\bullet \quad \mathbb{E} [\hat{\mu}_{\text{LHS},n}] = \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}[f(X_i)] = \frac{1}{n} \sum_{i=0}^{n-1} \int_{i/n}^{(i+1)/n} f(x) \times n \times dx = \int_0^1 f(x) dx = \mu$$

which means that $\hat{\mu}_{\text{LHS},n}$ is **unbiased**

- The X_i are **not IID** but they are **uncorrelated**



Variance of LHS in One Dimension

$$\text{var}(\hat{\mu}_{\text{LHS},n}) = \text{var}\left(\sum_{i=0}^{n-1} \frac{f(X_i)}{n}\right) = \frac{1}{n^2} \sum_{i=0}^{n-1} \text{var}(f(X_i)) \quad \text{since the } X_i \text{ are uncorrelated}$$

$$= \frac{1}{n^2} \sum_{i=0}^{n-1} \int_{i/n}^{(i+1)/n} (f(x) - f(\eta_i))^2 \times n \times dx$$

where $f(\eta_i) = \mathbb{E}[f(X_i)]$ for some η_i by the Mean Value Theorem

$$= \frac{1}{n} \sum_{i=0}^{n-1} \int_{i/n}^{(i+1)/n} (f'(\xi_i(x))(x - \eta_i))^2 dx,$$

where $f(x) = f(\eta_i) + f'(\xi_i(x))(x - \eta_i)$ for some $\xi(x)$ by Taylor's Theorem

$$\leq \frac{1}{n} \sum_{i=0}^{n-1} \frac{\|f'\|_{\infty}^2}{n^3} = \frac{\|f'\|_{\infty}^2}{n^3}$$

$$\text{rmse}(\hat{\mu}_{\text{LHS},n}) \leq \frac{\|f'\|_{\infty}}{n^{3/2}}$$